

RS004 Week 13 — English Worksheet

Topic: Enemies & Game Over · **Course:** Platformer Game · **Time:** about 45 minutes

This week is about **enemies**. You will read and think about an **Enemy class** that patrols left and right on its own, how touching an enemy triggers **Game Over**, and how pressing **R** restarts. A **game_state** variable decides what the game is doing. Then you will explain the code you wrote in today's live lesson.

Keep these words handy: **class**, **patrol**, **game_state**, **Game Over**, **restart (R)**.

1 · Predict 🧠

Read each snippet. Write what you think it does.

```
def update(self):
    self.rect.x = self.rect.x + self.direction * self.speed
    if self.rect.x > self.start_x + self.range_x:
        self.direction = -1
    if self.rect.x < self.start_x:
        self.direction = 1
```

The enemy reaches the right end of its range. What happens to **direction** ?


```
if player.colliderect(enemy.rect):
    game_state = "gameover"
```

The player touches an enemy. What does the game switch to?


```
elif game_state == "gameover":  
    if keys[pygame.K_r]:  
        player.x = start_x  
        player.y = start_y  
        game_state = "playing"
```

The game is over and R is pressed. What resets, and what does the game become?

2 · Read the Class

An `Enemy` is created with `Enemy(300, 530, 40, 40, 200)` . Match each value.

1. 300 2. 530 3. 40 4. 40 5. 200

A. patrol range (how far it walks)
B. start x
C. start y
D. width
E. height

1 → __ 2 → __ 3 → __ 4 → __ 5 → __

3 · Spot the Bug 🐛

Each snippet is broken. Rewrite it so it works, then explain why the original was wrong.

Bug A — The enemy should walk back and forth. Right now it walks off the right side and never comes back.

```
def update(self):  
    self.rect.x = self.rect.x + self.speed
```

Hint: there is no `direction`, so it only ever moves one way. It needs to flip at the edges.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — Movement should stop once the game is over. Right now the player keeps moving on the Game Over screen.

```
keys = pygame.key.get_pressed()  
if keys[pygame.K_LEFT]:  
    player.x -= PLAYER_SPEED  
if keys[pygame.K_RIGHT]:  
    player.x += PLAYER_SPEED  
for enemy in enemies:  
    enemy.update()  
    if player.colliderect(enemy.rect):  
        game_state = "gameover"
```

Hint: wrap the play logic so it only runs when `game_state == "playing"`.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — R should restart the game, but pressing it does nothing.

```
if game_state == "gameover":  
    player.x = start_x  
    player.y = start_y  
    game_state = "playing"
```

Hint: this restarts every frame with no key check. Only restart when R is pressed.

Write the fixed code:

Why was it wrong? Why does your fix work?

4 · Explain the Code

Here is a working enemy setup, just like the one from class. Read it carefully, then answer the questions.

```
class Enemy:
    def __init__(self, start_x, start_y, width, height, range_x):
        self.rect = pygame.Rect(start_x, start_y, width, height)
        self.start_x = start_x
        self.range_x = range_x
        self.direction = 1
        self.speed = 2

    def update(self):
        self.rect.x = self.rect.x + self.direction * self.speed
        if self.rect.x > self.start_x + self.range_x:
            self.direction = -1
        if self.rect.x < self.start_x:
            self.direction = 1

enemies = [Enemy(300, 530, 40, 40, 200)]

if game_state == "playing":
    for enemy in enemies:
        enemy.update()
        if player.collidect(enemy.rect):
            game_state = "gameover"
```

Which line saves where the enemy started, and why does the patrol need it?

What does `self.direction = 1` mean, and what is happening when it becomes `-1`?

Why does every enemy go inside the `for enemy in enemies:` loop?

What turns the game into Game Over, and on which line does it happen?

Why is all of this wrapped inside `if game_state == "playing":` ?

5 · Explain Your Lesson Code 🎥

Explain the code **you** wrote in today's lesson. Make a short phone video walking through your own game. You may show it running. Try to use these words: **class**, **patrol**, **game_state**, **Game Over**, **restart**.

1. Show your `Enemy` class and read the `update` out loud.
2. Explain how `direction` flips so the enemy patrols back and forth.
3. Touch an enemy and show `game_state` turn to Game Over.
4. Press R and explain how the restart resets the game.

Write what you will say in your video. Plan it here before you record.

Submit 

Send this worksheet + a video explaining your lesson code to teacher on KakaoTalk.